Section
Industry Engagement

Measuring the Value of (Research) Data

Whitepaper on behalf of the Section Industry Engagement
of the National Research Data Infrastructure (NFDI)

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Executive Summary

Data is a unique economic asset and information resource with special economic characteristics. Across various industries, firms are nowadays collecting and using data to optimize their processes, create new value propositions for their customers, or sell and exchange their data with external partners. This whitepaper outlines how firms can *measure* the value of their data and data-related activities.

Three main approaches exist to measure the business value of data. First, the data value can be determined by measuring its market prices, costs, and future income prospects (market-based approach). Second, the data's estimated utility in economic and public benefits may be used to quantify its value (economic approach). Third, various data- and context-related parameters may be used to estimate the value of data in a specific setting (dimensional approach).

We further find that data that is explicitly used for research purposes differs in these valuation approaches. Here, the value of data is often determined by its academic impact, reuse, and contribution to scientific breakthroughs. The FAIR principles also provide a framework for enhancing the utility and value of research data in this setting, guiding the assessment based on findability, accessibility, interoperability, and reusability of the data.

Nevertheless, the lack of a common conceptual framework and the difficulty of predicting data's future applications and relevance make the value measurement of data still challenging in practice.

1. Introduction

In the era of digital transformation, a new kind of data-driven economy has emerged, characterized by the increasing datafication of human, social, political, and economic activities (Ciuriak 2018). Therefore, data is becoming a key economic asset that influences business performance and drives growth. The effective transformation of data into actionable insights opens new possibilities for competitive advantage (Chase 2013; Lambrecht and Tucker 2017).

However, organizations face significant challenges when proving the value of their data and AI-related investments (Aziza 2023). Entrenched legacy systems, fragmented technologies and teams, and limited innovation in partnership networks ultimately hinder organizations from effectively demonstrating the business value derived from data (Aziza 2023). Furthermore, measuring the value of data is proving more difficult than expected due to the lack of standardized metrics (Fleckenstein et al. 2023).

This whitepaper aims to summarize existing approaches for measuring the value of data, both in a business and in an academic context. Our goal is to provide industrial and academic organizations with more insights and guidance to understand the drivers of the value of data. We proceed as follows: first, we will provide some conceptual foundations about what economic characteristics and processes turn data valuable. Based on this, we then summarize different existing measurement approaches for the value of data in business and academic settings. Finally, we discuss joint contemporary challenges to measuring the value of data.

2. Conceptual Foundations

Before turning to value measurement approaches, this section should first clarify what “data” exactly is in the context of this whitepaper, which special economic characteristics data has, and how data can generally be turned into value.

2.1 Characteristics and Value Drivers of Data

There exist many definitions of the term "data", each of which differs by the context in which it is used. The Organisation for Economic Co-operation and Development (OECD 2021), for example, defines data as "recorded information in structured or unstructured formats, including text, images, sound, and video." This broad definition captures the essence of data as a record, which includes all volumes of digitized information housed on servers and other hardware devices.

Yet, for the purpose of assessing data's value, a more targeted definition is warranted. The System of National Accounts (2022) takes a step forward with a definition tailored to this context: "[Data is…] Information content that is produced by accessing and observing phenomena; and recording, organizing, and storing information elements from these phenomena in a digital format, which provide an economic benefit when used in productive activities". This definition highlights the process of producing data and its subsequent utility in driving economic benefits. Note that by this definition, data is thus different from information (i.e., processed and contextualized data; see Pipino, Lee, and Wang 2002) and knowledge (i.e., information processed by the human brain; see Liew 2007).

Data has special economic characteristics that make it different from other consumer/industrial goods or content. While there are some similarities to content (e.g., a similar high fixed and low variable cost structure), other characteristics are unique for data (e.g., significant economies of scope; see Table 1 for an overview).

<table>
<tr><th>Consumer / Industrial Goods
(e.g., apples / aluminum)</th><th>Content
(e.g., online news, movies)</th><th>Raw Data
(e.g., sensor data)</th></tr>
<tr><td>On average medium fixed costs, medium variable costs</td><td colspan="2">High fixed costs, low variable costs</td></tr>
<tr><td>Can only be consumed once</td><td colspan="2">Infinitely durable and reusable</td></tr>
<tr><td>Quality easy to access</td><td colspan="2">Quality difficult to assess beforehand</td></tr>
<tr><td>Only limited availability</td><td colspan="2">Infinitely available (no scarce resource)</td></tr>
<tr><td>Product is itself valuable</td><td>Value created by direct consumption</td><td>Value created through use of analytics software (only intermediate product)</td></tr>
<tr><td>Rival good (only exists once)</td><td>Non-rival (possibility to share with other peers)</td><td>Non-rival (multiple entities of a company can use it simultaneously for different commercial purposes)</td></tr>
<tr><td>Constant value</td><td>Relatively constant value</td><td>Value may depreciate over time, depending on characteristic of data</td></tr>
<tr><td>Low heterogeneity concerning value perception among users</td><td>Medium heterogeneity concerning value perception among users</td><td>High heterogeneity concerning value perception among users</td></tr>
<tr><td>Low economies of scope</td><td>Low economies of scope</td><td>Significant economies of scope (combining different data is valuable)</td></tr>
</table>

Table 1: Differences between Consumer & Industrial Goods, Content and Data
(based on Schmarzo 2020 and Shapiro and Varian 1998)

Furthermore, the quality of data is paramount in driving its value. Data quality is often defined as a multi-dimensional construct relative to the consumers' evaluation based on their subjective desired standards (Golder, Mitra, and Moorman 2012). Consequently, data quality is viewed as "fitness for use by data consumers"—how well the data fulfills the objectives of the consumer and aligns with their expectations (Wang and Strong 1996).

Wang and Strong (1996) propose an empirical approach that delineates fifteen data quality dimensions, further categorized into four groups:

1. Intrinsic Data Quality: Evaluates the degree to which data reflects reality. It encompasses dimensions such as believability, accuracy, objectivity, and reputation, thus asserting that high-quality data is not only factually correct but also assessed credible and reliable from the consumer's perspective.

2. **Contextual Data Quality**: Accounts for the user- and task-specific relevance of data. For instance, the granularity required for financial data might differ between managers and auditors. This category includes the dimensions of value-added, relevancy, timeliness, completeness, and the appropriate amount of data, indicating that higher scores translate to a better fit for the user's circumstances, enhancing subjective data quality.
3. **Representational Data Quality**: Concerns the clarity and comprehensibility of data, involving dimensions like interpretability, ease of understanding, representational consistency, and concise representation. Well-represented and comprehensible data invariably raises its quality and utility.
4. **Accessibility Data Quality**: Addresses the availability and security of data access for the consumer, including dimensions of accessibility and access security. Hence, quality data is expected to be not only valuable but also suitably presented and readily available to the data consumer.

Both the economic characteristics of data as well as data quality-related dimensions are contributing to the value of data[1].

2.2 Data Value Generation Process

The inherent characteristics of data (see section 2.1) are not necessarily valuable on their own. It is the synergy between these characteristics and a company's technical ability to explore datasets through algorithms and experimentation that unlocks valuable insights, uncovers profit-enhancing opportunities, and creates the basis for strategic action (Lambrecht and Tucker 2017). The value of data is therefore often linked to the analytics applied to it and the resulting insights that are translated into decisive action. Data in a dynamic organization, or in an organization that is made accessible to others, becomes a more valuable asset than the same data in a more static organization (Coyle and Manley 2024).

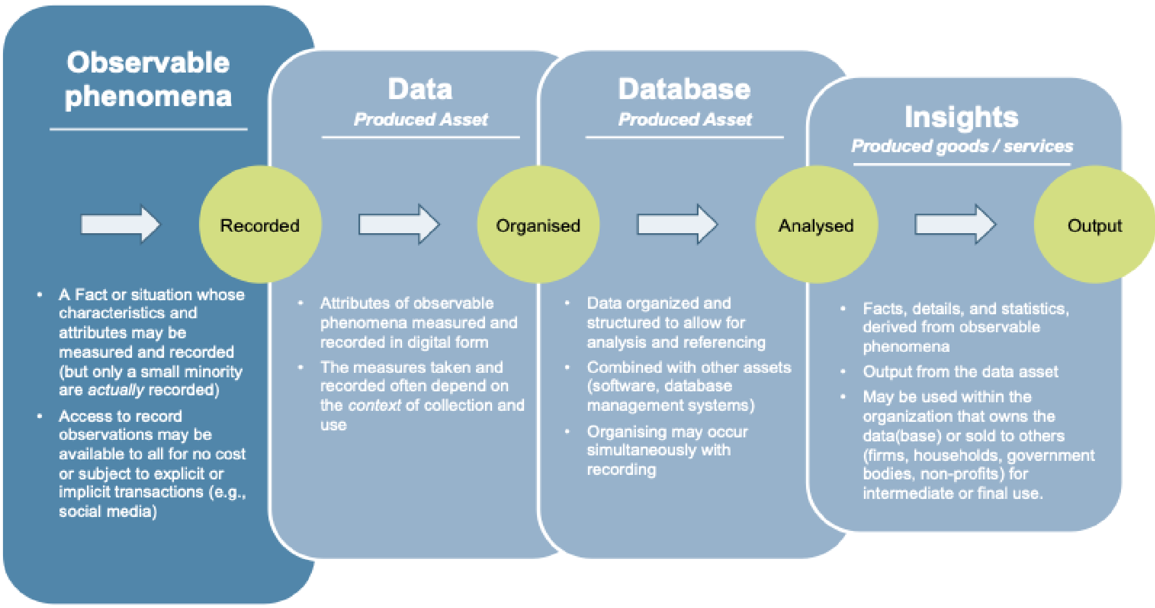

Figure 1: The Data-Information Chain (Mitchell et al. 2021)

[1] Note that metadata is also a highly important related concept to data. We have summarized some value drivers of metadata in the appendix.

Figure 1 outlines the transformation process from raw observations to actionable insights. Initially, observable phenomena are recorded - characteristics and attributes that are measurable and can be quantified. These attributes are then organized into a structured database, enabling analysis and reference, often combined with other systems. Through analysis, raw data is processed into meaningful facts, details, and statistics, termed as 'insights'. These insights are crucial end-products that can be utilized by the data's owners, such as firms or governments, to inform decisions and produce goods or services, thereby realizing the data's inherent value.

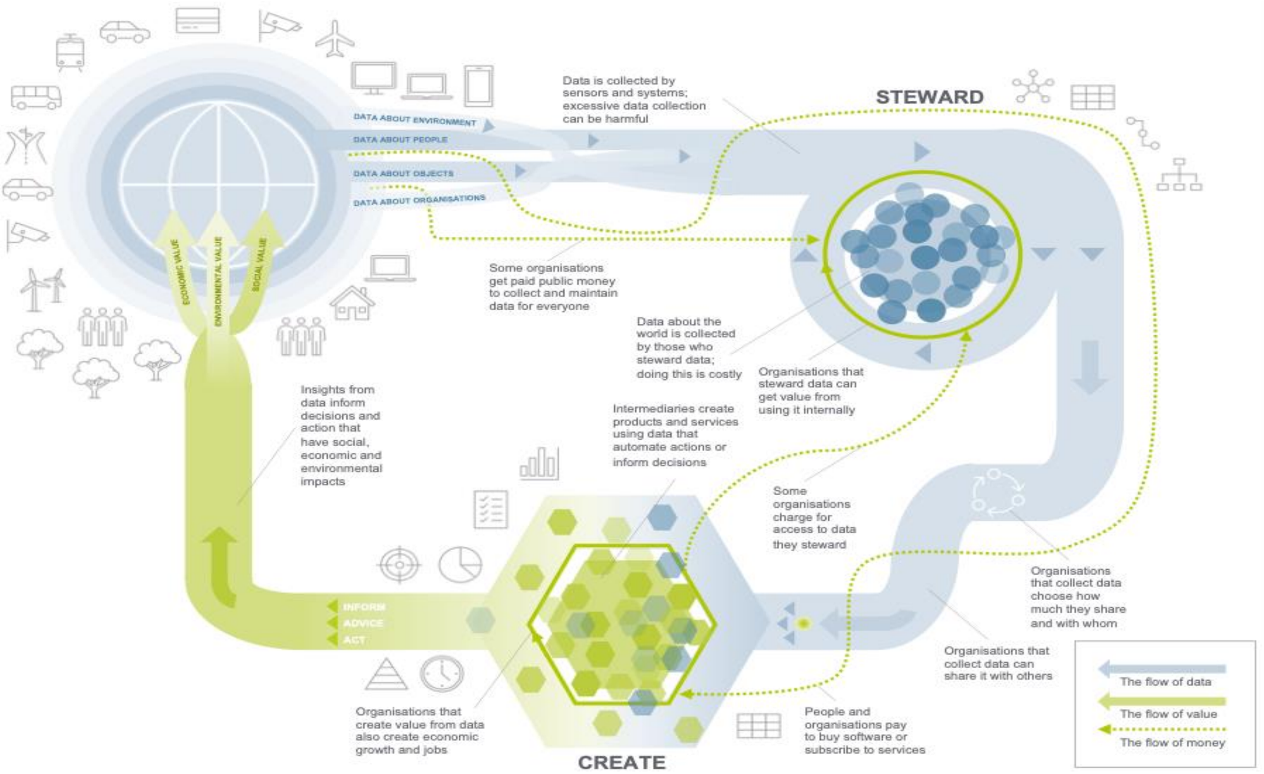

Figure 2: Data Value and Distribution (Coyle et al. 2020)

Figure 2 contains an alternative view on the value generation process through data. Here, data stewards help an organization to internally process data and create new offers that ultimately benefit the economy and create economic, environmental, and social value.

Overall, the combination of data's characteristics and technological expertise to turn data into insights can drive the value of data. In the subsequent section, we use these insights to understand how the created value can then be measured by the data owners.

3. Measuring the Value of Data

In this section, we review scientific and practitioner approaches that deal with data valuation. We split the approaches into measuring the value of business data and the value of academic value of research data[2]. Finally, we review measurement challenges that persist in both settings.

[2] Note that business data can also become academic research data, and vice versa (see Stahl, Hoff, and Hamann 2025).

3.1 Approaches to Measure the Business Value of Data

In the following, we draw on the approaches to (business) data valuation found by Fleckenstein et al. (2023). They argue that most approaches to data valuation typically fall into three categories: market-based valuation, economic models, and applying dimensions to data. Each model provides a different lens through which to evaluate the value of data, which will be explained in the following.

3.1.1 Market-Based Approach

The market-based approach defines the value of data in terms of cost and revenue for the firm. It can be differentiated into stock market-, cost-, and income-based valuations.

1. **Stock market-based valuations**

Stock market-based valuations look at observable market prices of data, if available. Highways England, for instance, analyzed its £115 billion in intangible assets to determine how much was attributable to data. By mapping data assets to business functions and estimating their potential market value, Highways England concluded that its data was worth £60 billion (Laney 2021). However, these market prices are not always available or can be confidently estimated. Furthermore, they may represent only a portion of data's total social value (Coyle and Manley 2024).

2. **Cost-based valuations**

Cost-based valuations represent the costs of generating and maintaining data (Coyle et al. 2020). The sum-of-cost approach adds up production costs such as: “intermediate consumption, compensation of employees, consumption of fixed capital used in production, a net return to fixed capital used in production (also known as "mark-up") and taxes (minus subsidies) on production” (OECD 2022). However, it may not capture the economic value of data assets fully, given data's non-decreasing returns, externalities, and the potential for monopolistic competition.

According to the OECD (2022), the sum-of-costs method appears to be the most viable for assessing the worth of own-account data, which is data produced internally by a firm not for the purpose of selling but for its own utilization. It also aligns with the valuation for other self-produced intellectual property products, such as software, and research and development initiatives. Efforts to formulate international guidelines for applying this method within the System of National Accounts are currently in progress (Coyle and Manley 2024; OECD 2022).

3. **Income-based valuations**

Income-based valuations aim to estimate the potential future income from data. The Net Present Value (NPV) approach is used for this, but its application to data assets is challenging due to the difficulty in predicting future income streams. Income-based approaches require more judgment compared to cost-based methods since data costs are more definitive than potential revenue streams (Coyle and Manley 2024; OECD 2022).

3.1.2 Economic Approach

Economic models take a different perspective by examining the value of the data for the broader economic and public benefit. If the data was made publicly available, which benefits could it bring? The estimated monetary value of these benefits could then be the value attached to this data.

Fleckenstein et al. (2023) further distinguish between studies that explicitly value open data and those that consider how policy contributes to public data value. Examples include the economic value of earth observation for Australia's economy and the financial implications of the California Consumer Privacy Act (2018) on business data valuation. Further explored by Coyle et al. (2020), economic models also consider the value generated for organizations that steward data, intermediaries that create products and services using data, and end-users who benefit from informed decisions and actions.

A major example is the provision of open data from Transport for London which generated significant economic and employment benefits. By making transportation data openly accessible, including real-time updates on schedules and service disruptions, the company enabled the creation of over 600 apps used by millions of commuters. A study by Deloitte (2017) estimated that this open data initiative contributes approximately £130 million annually to the economy. These benefits arise from improved travel efficiency, reduced congestion, and job creation in app development.

3.1.3 Dimensional Approach

Dimensional models yet take on a different approach by assessing the available data along certain self-defined properties of data (e.g., its data quality dimensions).

For instance, Fleckenstein et al. (2023) introduce a dimensional valuation model that encapsulates ownership, cost, usage, age, privacy, data quality, and volume and variety (see Table 2). Each dimension contributes to the overall assessment of a data's value, offering a more granular and multifaceted approach to valuation.

Dimension	Description
Ownership	Addresses outright data set ownership + licensing restrictions + service agreements
Cost	Addresses the cost of dataset acquisition, maintenance, and replacement
Usage	Addresses dataset mission critically, ability to integrate, usage scope, usage frequency, metadata, additional resources, expected increase in demand, and diminishing value
Age	Addresses refresh rate and available history
Privacy	Addresses whether the dataset contains sensitive data such as Personally Identifiable Information (PII), and meets privacy standards
Data quality	Addresses completeness, accuracy, currency, consistency, duplication, trustworthiness, and timelines (also see section 2.1)
Volume and variety	Addresses the number of records, scope of information for each record, and ability to answer needed questions

Table 2: Data Valuation Dimensions (Fleckenstein et al. 2023)

3.2 Approaches to Measure the Academic Value of Research Data

Research data differs from general data in its purpose. It is collected, observed, or used to answer key research questions or to test specific hypotheses. Despite the crucial role of research data, a standardized framework for its assessment remains elusive. Below, we present two types of directions according to which the value of research data can potentially be assessed and measured.

3.2.1 The FAIR Principles

The FAIR principles—findability, accessibility, interoperability, and reusability—serve as a guide for improving the utility and value of research data. The following metrics based on the FAIR principles can be defined and used to quantify the value of research data:

1. **Findability** can be achieved by, e.g., assigning unique identifiers (DOIs) and following a metadata schema. Researchers can then measure the *number of accesses* as one indicator of value for the academic community.
2. **Accessibility** can be assessed by the ease with which the data can be obtained and used by researchers. *The number of downloads of a given dataset* may be one measurable indicator of value here.
3. **Interoperability** describes by how well the data can be linked to other datasets. Datasets that are compatible with various platforms and systems are more versatile and, therefore, more valuable. It can, for instance, be measured by the *extent of adherence to existing standards and framework*s that then enable integration for others.
4. **Reusability** refers to how well-documented and structured the data is for further use. It can, therefore, be assessed by the *thoroughness of the data documentation and the robustness of its structure. The scientific impact generated by reusing this data* (e.g., number of research papers using the data in top publications) may also be used as a value measure here.

Adherence to these principles ensures that data is findable and usable and can be seamlessly integrated into ongoing research, increasing its value (GO FAIR 2025; Wilkinson et al. 2016). In addition, the European Commission measured the cost of not having FAIR research data. They found out that the measurable minimum cost of not having FAIR research data makes €10.2bn per year in Europe (European Commission 2018).

3.2.2 Societal and Economic Impact Measures

The societal and economic impact of research data is a crucial dimension in measuring its value, as it extends beyond academic circles to influence public policies, societal outcomes, and economic growth. It addressed the broader utility of data in addressing complex global challenges and fostering innovation.

On the one hand, research data may impact policymaking. Data-driven insights have the potential to inform and shape public policies and regulations. For example, environmental data plays a critical role in crafting climate change policies by providing evidence on emissions, biodiversity loss, and the efficacy of mitigation strategies. Similarly, health data has been instrumental in guiding medical guidelines, public health initiatives, and responses to pandemics, demonstrating the direct link between research data and societal well-being. Measuring the *monetary value of these public policy effects* may potentially also be linked to the value of data that made these effects possible.

On the other hand, the generated research data may also contribute to insights which can then help *firms save costs and/or increase their revenues*. By enabling more efficient resource allocation and driving technological innovation, research data then supports economic activities and growth. For example, transportation data has been used to reduce congestion and improve public transit systems, yielding significant savings and increased productivity. Similarly, data and insights from agricultural research has driven innovations in farming practices, enhancing yields and contributing to food security.

Together, the FAIR principles and societal and economic value measures highlight the extensive value of research data.

3.3 Measurement Challenges

According to Bendechache et al. (2023), both researchers and organizations face some major challenges in measuring the business and academic value of data.

First, the lack of a common conceptual framework is a significant hurdle. Data is ubiquitous and diverse, which means that its value can be subjective and context-dependent. Without a universally accepted framework that delineates clear definitions and standards for measuring data's value, assessments can vary widely, leading to inconsistency and confusion.

Second, predicting future applications and therefore the future value of data presents a unique challenge. Data that may seem of limited value today can become invaluable tomorrow if new uses are discovered. Conversely, data deemed valuable now might become obsolete as technologies and societal needs evolve. This temporal aspect of data valuation demands that any assessment method considers the potential for future utility and relevance.

4. Conclusion

The importance of data to gain a competitive advantage is ever-increasing in today's economy. Various firms are currently investing in data- and AI-related initiatives (Aziza 2023). Nevertheless, proving the value of data usage within the company is often a difficult endeavor. This whitepaper shows that firms can take on different perspectives to measure the value of their data. That is, market-based, economic, and dimensional approaches exist to model the value of data. In an academic setting, FAIR framework can be used as an additional value assessment benchmark.

Future research endeavors may focus on developing standardized frameworks and methodologies to facilitate more accurate and transparent data valuation practices. Additionally, advancements in technology, such as machine learning and (generative) artificial intelligence, hold promise for enhancing the precision and scalability of data valuation models.

Appendix: Metadata and its Value Drivers

Companies harness practical knowledge about their customers and business strategies to bolster decision-making. Achieving such objectives mandates a robust data management policy, which is incomplete without considering metadata management. Metadata is not an additional component, but an essential part of the structure and usefulness of the data itself (Elouataoui et al. 2022).

The most common definition of metadata is "data about data". This definition, while accurate, belies the depth and importance of metadata. Its ubiquitous nature and critical role in modern digital interactions means it is far more than just a subset of data. It is a “statement about a potentially informative object” (Pomerantz 2015). Metadata is always structured to a certain degree. It is collected in such a way that it can serve a useful purpose and organized into known categories. This concept of structure is what transforms raw information into usable metadata (Riley 2017). Metadata is pervasive in the contemporary landscape, integral to nearly every electronic device we interact with. Yet, when effectively implemented, metadata recedes into the background, becoming an unnoticed framework that enhances the user experience and system functionality (Pomerantz 2015).

Metadata and data have four key differences (Atlan 2023):

1. Use: One of the primary uses of data is to provide raw, unprocessed insights and to reveal hidden patterns that inform decision-making. Metadata, on the other hand, serves to bring comprehensive understanding to the data, offering the context and reference needed to make sense of these raw insights.
2. Value: Data may fluctuate in its usefulness and relevance. Some data may prove invaluable, driving critical decisions or innovations, while other data may remain dormant, never quite finding a practical application. In contrast, metadata is invariably valuable because it consistently enhances the organization, accessibility, and interpretability of data. It ensures that data is not just a collection of potential information but a well-documented resource that can be effectively employed when needed.
3. Processing: Data can be collected and stored in its raw form, to be processed later as required. Metadata, however, is inherently processed information; it is organized and categorized from the outset to ensure that it fulfills its role in data management and utilization effectively.
4. Management: The management of data is often bespoke, tailored to the specific type of data and its intended use case. Different types of data may demand different storage solutions, management policies, and usage protocols. Conversely, metadata management can be streamlined and generalized across an organization. The same metadata principles and systems can be applied universally, irrespective of the varying types of data that an enterprise may handle.

The value drivers for metadata hardly differ from the value drivers for data. Their value is largely determined by the quality of the metadata. Maintaining the quality of metadata is crucial as it has a direct impact on the integrity of data and the dependability of the insights derived. In Appendix A1, we summarize the conceptualization of Elouataoui et al. (2022) who categorize the key metadata quality dimensions into four main areas: usability, reliability, availability, and interoperability.

Quality Aspect	Metadata Quality Dimension	Meaning
Usability	Completeness	Assures that there are no missing attributes, and all the expected attributes have values.
	Consistency	The extent to which metadata is coherent and aligned with metadata standards and schemas.
	Usefulness	The extent to which metadata is valuable and relevant for its intended use.
Reliability	Accuracy	The degree to which metadata is correct and reliable.
	Timeliness	Refers to how recent and up-to-date the metadata is.
Availability	Accessibility	The extent to which metadata is available and easily accessible.
	Security	It consists of ensuring that access to metadata is appropriately restricted.
Interoperability	Shareability	The extent to which metadata could be effectively used out of its local environment.
	Discoverability	The extent to which metadata is visible and can be easily found.
	Extendibility	The extent to which a new version may be easily extended.
	Versionability	The extent to which a new version may be easily created.

Table A1: Metadata Quality Dimensions (Elouataoui et al. 2022)

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